

Practice Sheet

Chapter 3 (ER Models)

***Solutions are at the end of the document. SOLVE IT YOURSELF first.**

Question 1: Suppose you are given the following requirements for a simple database for the National Hockey League (NHL):

- The NHL has many teams,
- Each team has a unique name, a city and a coach
- Each player belongs to only one team.
- Each player has a unique phone number, name, a position (such as left wing or goalie), a skill level, and a set of injury records.
- Each team has exactly 1 captain who is a player.
- a game is played between two teams (referred to as host_team and guest_team) and has a date (such as May 11th, 1999) and a score (such as 4 to 2).

Construct a clean and concise ER diagram for the NHL database.

Question 2: Draw the ER diagram which models an online bookstore.

- Books have year, title, price, unique ISBN and at least 1 author. Books are published by exactly 1 publisher. Each publisher has a unique name, address, phone, email and unique website.
- Customers have a unique userID, name, phone number, email and billing address. Billing address is composed of street, city, zip, country.
- Each customer has one shopping cart, the shopping cart has a unique id. Books can be added to shopping carts. They can add more than one book of the same title at the same time, so quantity and date attributes are required.
- All copies of the same book are stored in one of the warehouses owned by the bookstore; The warehouse has a unique id, an address, a phone number and a manager.

Question 3: A pet store wants to design a database system for inventory and sales tracking. The requirements for the store are given below:

- The store sells different kinds of pets. The pets have a name, type, description, unique tag number, price and cost.
- The profit for each pet is calculated using price and cost. Profit will not be stored in the database but should be shown with the appropriate symbol in the diagram.
- Different suppliers supply the pets to the store. The supplier name, location and unique phone numbers are stored. Also, the shipping time and date from the supplier for each pet is recorded.
- The store keeps a record of all customers- national_id, name, address, email and contact. The customers can buy 1 or more pets from the store. The delivery date and time for the purchase is also recorded.
- The pets are often taken to the Vet in case of diseases or checkups. The vet's address, name, unique phone and fees are stored. For each visit to the vet, the date, time, all medicines and reason are recorded.